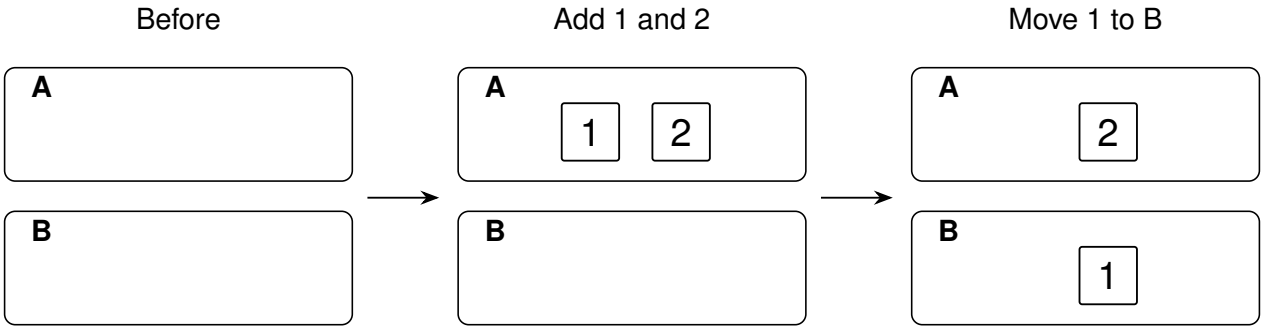
Both bowls start empty. Each day, put the next two numbered cards in A, then move A's smallest card to B. Cards in B stay there.



Problem 1: Run six days with cards 1–12. Can the bowls ever end a day with different numbers of cards? You may continue to twelve days with cards 1–24.

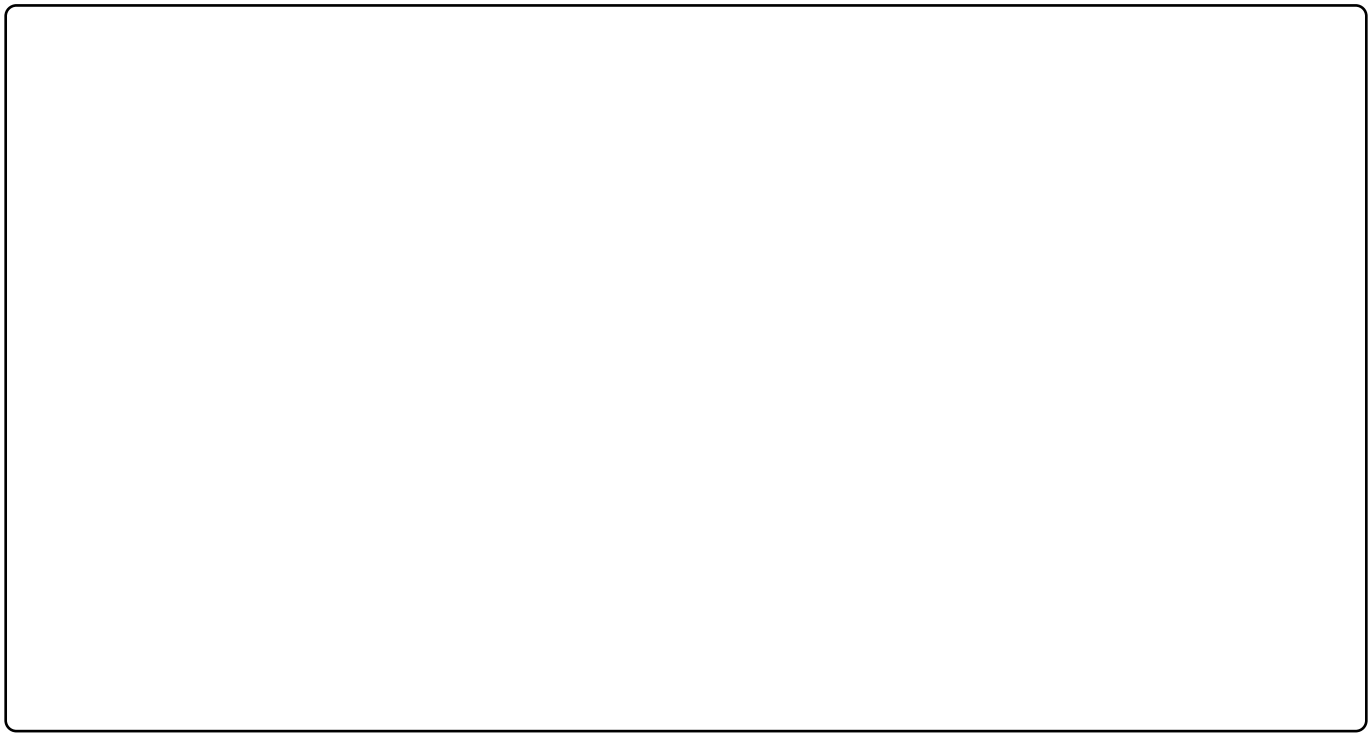
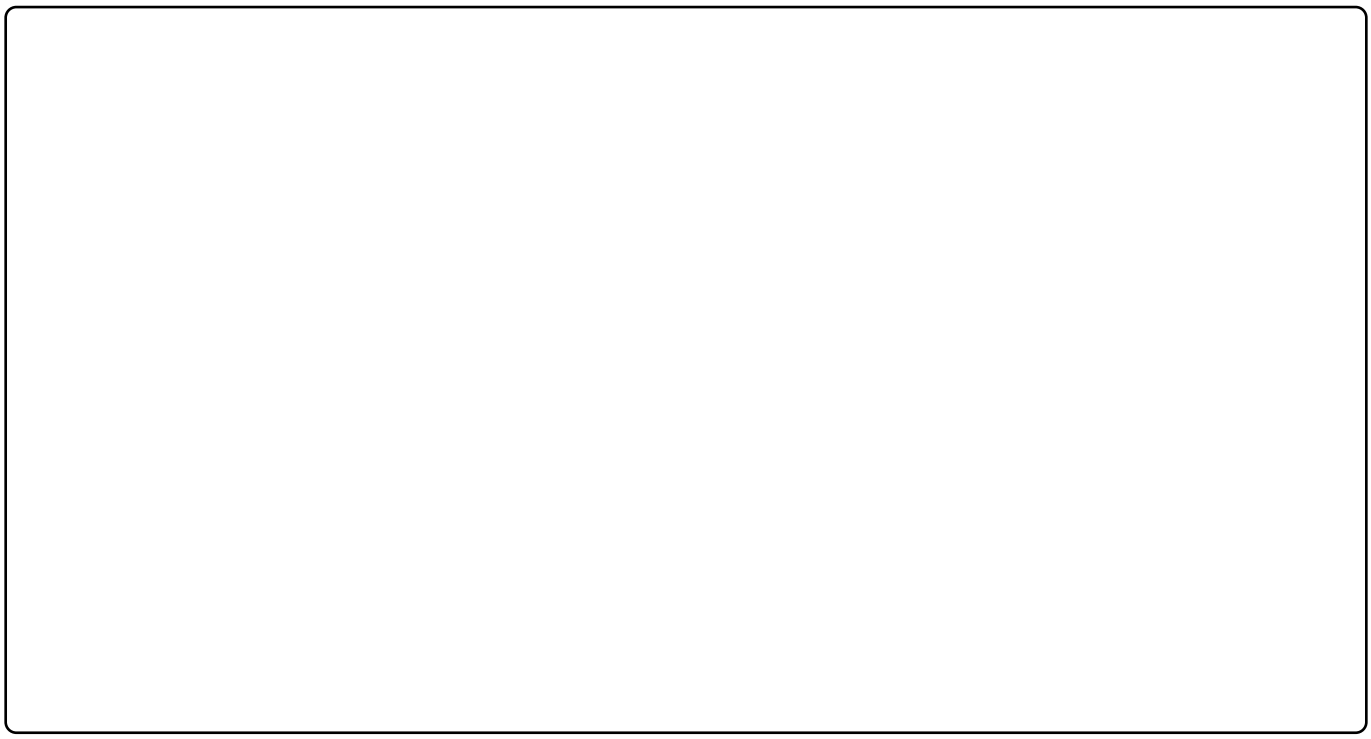
Problem 2: Restart. Find a rule for the day any numbered card moves to B. Test your rule on cards 3, 6, and 10, and predict the day for card 20 without running twenty days.

3	6	10	20
Day _____	Day _____	Day _____	Day _____

Problem 3: Start again. Each day, move the larger of today's two new cards to B, instead of A's smallest card. Could somebody tell which rule you used from the bowl counts alone? Find a way to show the difference using the cards.

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24

A



B